

ShrFlow Email Engine

How the AI Chatbot Works Without External API Keys

1. The Offline Architecture Concept

Standard conversational chatbot applications connect to external APIs (like OpenAI ChatGPT or Google Gemini) to understand user requests. However, these external requests create security risks, increase latency, and charge API usage fees. To solve this for self-hosted instances, ShrFlow features an offline **Natural Language Processing (NLP) state-machine**.

Rather than trying to understand the entirety of human language, the bot is purpose-built to navigate your workspace databases. It intercepts user queries locally, extracts targeted intent tokens (keywords), queries the local PostgreSQL database, and formats a real-time markdown response.

2. Technical Execution Flow

The backend chatbot processes queries in four distinct phases:

- 1. Live Metric Querying:** The backend queries the database for the active tenant's current contacts, campaigns, sent messages, open rates, and bounces.
- 2. Query Normalization & Token Matching:** The query is lowercased. The backend route scans for key vocabulary words (tokens): 'open', 'click', 'bounce', 'contact', 'subscriber', 'campaign'.
- 3. Intent Redirection:** The matching token determines which code block handles the query.
- 4. Rendered Markdown Response:** The code dynamically inserts the real-time calculated values from Step 1 into a pre-designed string template, along with optimization advice.

3. Concrete Example Execution Trace

Here is exactly how a request is processed under the hood:

Step	Action / Code Logic
User Query	User types: <i>'Can you tell me about my bounce rate?'</i>
Database Fetch	API queries the database: • num_sent = 1,000 • num_failed = 10 • num_bounces = 25 • calculated bounce_rate = 2.48%

Token Parsing	Query converted to lowercase. The backend scans string for intent: if 'bounce' in query: -> Match Found!
Template Interpolation	Values are plugged into template: <i>'You have recorded 25 total bounces, yielding a bounce rate of 2.48%...'</i>
JSON Output	FastAPI returns response: {'response': '...'} in less than 50 milliseconds .

4. Key Product Advantages

- **Zero Cost:** Requires no subscriptions or pay-per-token API accounts.
- **Offline Support:** Runs locally within Docker containers without internet dependencies.
- **Absolute Data Privacy:** Senders' metrics are never transmitted to third-party services.

5. Developer Directory References

Backend API route handler: `platform/api/routes/analytics.py` (functions: `ai_chat`)

Frontend React Component: `platform/client/src/components/analytics/AIChatbot.tsx`